

CropSense

AI-Powered Pre-Symptomatic Disease Intelligence

[VERSION 2.0 | WORKFLOW & ARCHITECTURE UPDATE]



The 7-Day Blindspot



Invisible Infection: Pathogens divide at the cellular level with ZERO visual surface signs for the first 7 days.



The Damage Lag: By the time spots appear, 40-60% of plant tissue is already destroyed.



Treatment Gap: Traditional labs take 10+ days—far too late to contain outbreaks.

Infected leaf at microscopic scale



Economic & Social Impact

₹90k

Crore Annual Loss (India)

A Crisis for Smallholders

Over **146 Million** farmers lack the capital to absorb repetitive crop failures. With only **15%** access to experts, the "Wait and See" approach leads to total field devastation.

CropSense provides the **Proactive Shield** needed to safeguard food security.

The Solution Architecture

Combining NIR Spectroscopy with Edge Intelligence.

NIR Spectroscopy Sensing

Light-Based Diagnosis

CropSense uses dual-wavelength light penetration to "see" through leaf tissue:

- ⚡ **850nm NIR:** Detects internal structural changes.
- 🌿 **660nm Red:** Measures chlorophyll concentration.
- 🔧 **Accuracy:** 92% classification via 16-bit signal conversion.



System Workflow Architecture

02. Digitization

The ADS1115 16-bit ADC converts raw light intensity into high-resolution digital data vectors.

04. Output Action

SIM800L triggers an SMS alert with localized dosages. OLED and Buzzer provide instant field feedback.

01. Input Sensing

Optical leaf-clip pulses NIR and Red light. Photodiodes capture the light passing through the tissue.

03. Edge Inference

ESP32 runs the TinyML model locally. The AI classifies the signature against known disease profiles.

"A seamless flow from physical leaf-capture to actionable digital remedy."

Competitive Benchmarking

Detection Window (Days Early)

14 Days (CropSense)

Mobile Camera Apps

0 Days (Reactive)

Lab Tissue Testing

7+ Day Delay

Unlike visual AI, CropSense is **proactive**, not reactive.

1-Acre Scalability Strategy

The Sentinel Grid

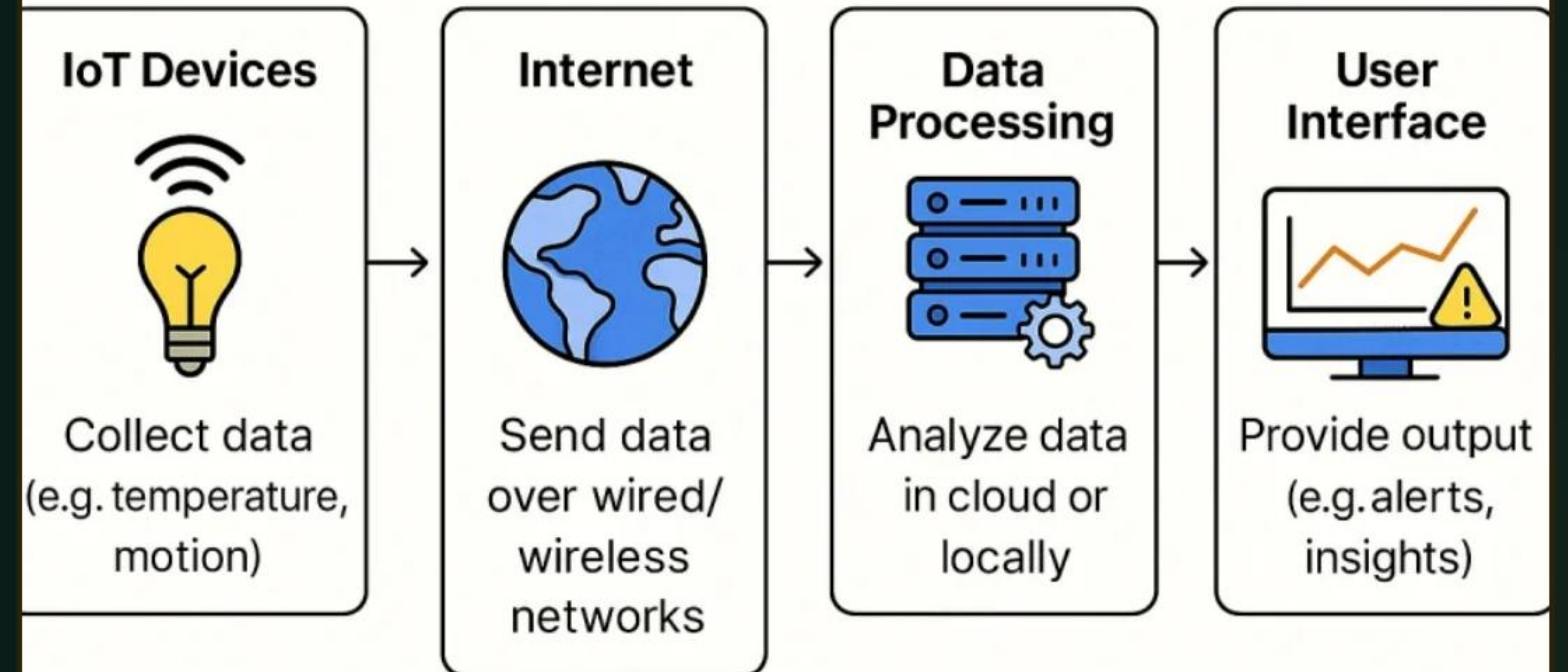
Protecting 5,000 crops with just one device:

Strategic Sampling: Scan 5 "Sentinel Plants" at windward corners and low-lying zones.

Early Warning: One scanning session per day covers the whole field.

Efficiency: ₹2,000 protects ₹50,000+ in crop value.

How IoT Works



The Tech Stack



TinyML Edge

Quantized neural networks running on ESP32. 100% Offline inference.



GSM Alerting

SIM800L module bypasses internet needs to reach any basic phone.



Solar Self-Power

6V Solar Panel + Li-Po ensures operation in zero-grid zones.

Global Scaling Roadmap



Mass Production

Transitioning from dev-boards to SMT custom PCBs to drop unit cost by 40%.



NB-IoT Upgrade

Upgrading to Global IoT bands for deployment in USA, EU, and Asia.



B2B2C Sales

Partnering with Insurance & Fertilizer giants for mass distribution.

The Future of Agriculture

A Health-First Ecosystem

Our vision is to replace "Pesticide Overuse" with "Precision Health Management". By catching disease early, we reduce chemical reliance by 40% and boost yield by 25%.

CropSense: Protecting the world's food security, one leaf at a time.



Thank You

Questions? We'd love to hear your feedback.

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